CLASS 8 — SCIENCE · FORMULA SHEET

Chapter 6: **Pressure, Winds, Storms and Cyclones** | Quick revision, one page

Pressure

- $P = \dfrac{F}{A}$; force acts perpendicular to the surface.
- SI unit: pascal (Pa); $1\ \text{Pa} = 1\ \text{N/m}^2$.
- Same force: smaller area gives larger pressure (knife, nail); larger area gives smaller pressure (broad straps, tractor tyres).

Liquids and gases

- Liquids exert pressure on the base and walls, in all directions.
- Liquid pressure increases with depth.
- Air has weight and exerts atmospheric pressure; it decreases with height.
- Gases also exert pressure in all directions on the walls of a container.

Winds

- Air moves from high pressure to low pressure; this moving air is wind.
- Uneven heating of land and sea (sea breeze, land breeze) causes pressure differences.
- Warm air rises, leaving a low-pressure region.

Storms and cyclones

- Thunderstorm, tornado and tropical cyclone: very low-pressure systems with strong winds.
- Cyclones form over warm oceans; the eye is the calm centre.
- Names: hurricane (Atlantic), typhoon (Pacific), cyclone (Indian Ocean).
- Safety: early warning, shelters, evacuation, mangroves/trees as barriers; storm surge causes coastal flooding.

MASTERMIND